

Non-density as a Brauer–Manin obstruction

Why $E(\mathbb{Q})$ misses an open set of $E(\mathbb{Q}_p)$: the local Tate pairing and reciprocity, carried by an explicit quaternion algebra on E — and the exact point where quaternion algebras run out

companion to ec-padic-closure.typ and integral-bm.typ; checked in ec-density-bm.gp, output in results/ec-density-bm.txt

Yes — and the two are one argument. On an elliptic curve the Brauer–Manin pairing **is** the sum over places of the local Tate duality pairing, and reciprocity is global duality. Made explicit through a 2-isogeny it becomes a Hilbert symbol. On

$$E : y^2 = x^3 - 2x^2 + 2x \quad (\text{conductor } 128, \text{rank } 1),$$

the quaternion algebra $\mathcal{A} = (7, x) \in \text{Br}(E)$ has $\text{inv}_v \mathcal{A} = 0$ at every place $v \neq 7$ and every local point; Hilbert reciprocity then forces $\text{inv}_7 \mathcal{A} = 0$ at every **rational** point. But $\text{inv}_7 \mathcal{A} = 1/2$ at the point of $E(\mathbb{Q}_7)$ with $x = 3$. That open set contains no rational point — which is exactly the index-2 non-density at $p = 7$.

1. The general statement

Let $E/\mathbb{Q}$ be an elliptic curve. Because E has a rational point,

$$\text{Br}_1(E)/\text{Br}_0(E) \cong H^1(\mathbb{Q}, \text{Pic } \overline{E}) \cong H^1(\mathbb{Q}, E),$$

the Weil–Châtelet group. For $\beta \in H^1(\mathbb{Q}, E)$ with associated Brauer class $\mathcal{A}_\beta$, the invariant at a local point is the **local Tate duality pairing**

$$\text{inv}_v \mathcal{A}_\beta(Q_v) = \langle Q_v, \beta_v \rangle_v, \quad \langle \cdot, \cdot \rangle_v : E(\mathbb{Q}_v) \times H^1(\mathbb{Q}_v, E) \longrightarrow \mathbb{Q}/\mathbb{Z},$$

a perfect pairing of a profinite group with a discrete torsion one. The Brauer–Manin condition $\sum_v \text{inv}_v = 0$ is then literally

$$\sum_v \langle Q_v, \beta_v \rangle_v = 0,$$

which for a **global** point is the Poitou–Tate/Cassels global duality relation. So the two phrasings in the question are the same argument in two notations.

Manin’s theorem. If $\text{III}(E)$ is finite, then the Brauer–Manin set of E is **exactly** the closure of $E(\mathbb{Q})$ in the adelic points:

$$E(\mathbf{A}_{\mathbb{Q}})_{\bullet}^{\text{Br}} = \overline{E(\mathbb{Q})}.$$

So every failure of density — at one place or adelically — is a Brauer–Manin obstruction, and nothing else is. Both curves below have analytic rank 1 (checked with `ellanalyticrank`), so Gross–Zagier and Kolyvagin make III finite and the theorem applies to them unconditionally.

That settles the question in principle. The content is in **exhibiting the class**, and for that one wants an isogeny.

1.1. The pairing as a Hilbert symbol

Take $E : y^2 = x(x^2 + ax + b)$ with the rational 2-torsion point $T = (0, 0)$, and let $\varphi : E \rightarrow E'$ be the 2-isogeny with kernel $\langle T \rangle$. The descent map

$$E(\mathbb{Q}_v)/\hat{\varphi}E'(\mathbb{Q}_v) \hookrightarrow \mathbb{Q}_v^\times/(\mathbb{Q}_v^\times)^2, \quad Q \mapsto x(Q) \quad (T \mapsto b, O \mapsto 1)$$

is injective, and the Tate pairing against the dual class $d \in \mathbb{Q}_v^\times/(\mathbb{Q}_v^\times)^2$ is the Hilbert symbol:

$$\langle Q, d \rangle_v = \frac{1}{2}(d, x(Q))_v.$$

The Brauer class. $\mathcal{A}_d = (d, x)$, a quaternion algebra over $\mathbb{Q}(E)$. It is **unramified** on E : for a constant d the tame symbol is $\partial_Q(d, f) = \bar{d}^{v_Q(f)}$, and

$$\text{div}(x) = 2(T) - 2(O)$$

has even multiplicities everywhere — x has a double zero at the 2-torsion point T and a double pole at O . So every residue is trivial and $\mathcal{A}_d \in \text{Br}(E)$.

Reciprocity for this class is Hilbert reciprocity, $\prod_v (d, x(Q))_v = 1$. The recipe is therefore: **find d such that $(d, x)_v$ is identically +1 on $E(\mathbb{Q}_v)$ for every $v \neq p$, and not identically +1 on $E(\mathbb{Q}_p)$.**

Two of the places are free, for reasons visible in the equation:

- **The real place.** If $a^2 - 4b < 0$ then $x^2 + ax + b > 0$, so $y^2 = x \cdot (\text{positive})$ forces $x \geq 0$ at every real point. Then $(d, x)_\infty = +1$ for **every** d .
- **Odd v of good reduction.** Every point has $v_v(x)$ even: if $v_v(x) > 0$ then $x^2 + ax + b \equiv b$ is a unit (good reduction at an odd v forces $v \nmid b$, since $v \mid b$ would divide the discriminant $16b^2(a^2 - 4b)$), so $v_v(y^2) = v_v(x)$; and $v_v(x) < 0$ gives $x = A/C^2$. So the local image consists of **unit** classes, and the Hilbert symbol of two units at an odd place is trivial whenever d is a v -adic unit.

What is left is $v = 2$, the odd bad primes, and the primes dividing d — a finite check.

2. Case 1: good reduction, conductor 128 at $p = 7$

$$E: y^2 = x^3 - 2x^2 + 2x, \quad \Delta = -256, \quad \text{rank } 1, \quad E(\mathbb{Q}) = \langle (1, 1) \rangle \oplus \langle (0, 0) \rangle \cong \mathbb{Z} \oplus \mathbb{Z}/2.$$

At $p = 7$ the reduction is good with $E(\mathbb{F}_7) \cong \mathbb{Z}/12$, and the filtration method of `ec-padic-closure.typ` gives closure index 2. Take $d = 7$, so $\mathcal{A} = (7, x)$.

place	$\text{inv}_v \mathcal{A}$	why
$v = \infty$	0	$x^2 - 2x + 2 = (x - 1)^2 + 1 > 0$, so $x \geq 0$ on $E(\mathbb{R})$
odd $v \neq 7$	0	$v_v(x)$ even, and 7 is a v -adic unit
$v = 2$	0	checked on 61 representatives of the local image
$v = 7$	varies	$(7, x)_7 = \left(\frac{x}{7}\right)$, and the image contains non-residues

So for $Q \in E(\mathbb{Q})$ all terms but $v = 7$ vanish, and $\prod_v (7, x(Q))_v = 1$ forces $(7, x(Q))_7 = +1$. Verified directly on nG and $nG + T$ for $n \leq 8$: the invariant is 0 at every place and the product is 1, as it must be.

And the class is not constant at 7. Take $x = 3$: then $x(x^2 - 2x + 2) = 3 \cdot 5 = 15$, which is a square in $\mathbb{Q}_7$ (it is a unit and $15 \equiv 1 \pmod{7}$). So $E(\mathbb{Q}_7)$ contains a point with $x = 3$, and

$$(7, 3)_7 = \left(\frac{3}{7}\right) = -1 \quad \text{since } 3 \text{ is a non-residue mod } 7.$$

Its 7-adic neighbourhood contains no rational point at all. Since $Q \mapsto (7, x(Q))_7$ is a homomorphism $E(\mathbb{Q}_7) \rightarrow \{\pm 1\}$ and is onto, its kernel has index exactly 2 — matching the index computed from the filtration. **The Brauer class explains the whole failure.**

3. Case 2: bad reduction, conductor 960 at $p = 3$

$$E : y^2 = x^3 - 4x^2 + 9x, \quad \Delta = -25920 = -2^6 \cdot 3^4 \cdot 5, \quad \text{rank } 1, \quad E(\mathbb{Q}) = \langle (4, 6) \rangle \oplus \langle (0, 0) \rangle.$$

At $p = 3$ the reduction is **bad**, of type I_4 , non-split ($a_3 = -1$), with $c_3 = 2$ and $|E(\mathbb{Q}_3)/E_1(\mathbb{Q}_3)| = c_3(p - a_p) = 8$; the closure has index 2. The singular point of the reduction mod 3 is $(0, 0)$, which is the 2-torsion point — so here $T \notin E_0(\mathbb{Q}_3)$, the image in the component group is full, and the failure sits in the torus layer E_0/E_1 .

Take $d = 3$, so $\mathcal{A} = (3, x)$. The same four lines run: $x^2 - 4x + 9$ has discriminant $-20 < 0$ so $x \geq 0$ on $E(\mathbb{R})$; odd $v \neq 3$ contribute nothing; $v = 2$ is checked directly; and at $v = 3$ the symbol $(3, x)_3 = (\frac{x}{3})$ varies. The point of $E(\mathbb{Q}_3)$ with $x = 2$ has $x(x^2 - 4x + 9) = 10$, a square in $\mathbb{Q}_3$, and $(3, 2)_3 = -1$.

Nothing in the argument cared whether p was a prime of good or of bad reduction. The Hilbert symbol does not know about Kodaira types; it only sees $x(Q)$ modulo squares. The filtration picture of `ec-padic-closure.typ` tells you **which layer** fails; the Brauer class tells you **why no rational point can be there**, and it is the same class in both cases.

4. Where quaternion algebras stop

A class of order 2 in $\text{Br}(E)$ cuts out a subgroup of index at most 2. So the explicit quaternion story reaches exactly the index-2 failures — and no further:

curve (conductor)	p	index	reachable by a quaternion class?
$y^2 + y = x^3 - x$ (37)	23	2	yes
$y^2 + y = x^3 + x^2$ (43)	43	2	yes
$y^2 + y = x^3 - x$ (37)	59	4	no — needs order 4
$y^2 + xy + y = x^3 + x^2 - x$ (89)	11	11	no — needs order 11

The conductor-89 example at $p = 11$ is the sharpest case: there the reduction map $E(\mathbb{Q}) \rightarrow E(\mathbb{F}_{11})$ is **surjective** and the failure is one layer deeper, in $E_1/E_2 \cong \mathbb{F}_{11}$, with index 11. No quaternion algebra can see it. Manin's theorem still applies — III is finite there — so a Brauer class of order 11 does the job, but it is not an algebra one writes down by hand.

The moral. Brauer–Manin explains all of it, by Manin's theorem. What the **explicit** quaternion construction reaches is the index-2 part, which is the reduction layer — and that is no accident: the pairing $\langle Q, d \rangle_v = \frac{1}{2}(d, x(Q))_v$ is exactly the local pairing of 2-descent, so the classes one can write down are precisely those that 2-descent already sees. Deeper layers need deeper descent.

5. Does an index-2 failure need a 2-isogeny? No

Section 2 and Section 3 both used a rational 2-torsion point, and it would be easy to read that as a hypothesis. It is not one — and the reading has a clean counterexample, namely the very first curve of `ec-padic-closure.typ`.

First, the terminology: a 2-isogeny **defined over $\mathbb{Q}$** is one with Galois-stable kernel of order 2, i.e. exactly a rational point of order 2, i.e. a rational root of the cubic. So the question is whether an index-2 failure forces $E[2](\mathbb{Q}) \neq 0$.

What is actually needed is a class of order 2 in $\text{Br}(E)$, equivalently a class in $H^1(\mathbb{Q}, E[2])$ — and full 2-descent on $E[2]$ exists for **every** elliptic curve. The isogeny in Section 2 was a convenience, not a hypothesis: a rational 2-torsion point puts the descent map into $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, so the pairing is a Hilbert symbol over $\mathbb{Q}$ and the Brauer class is a **single quaternion algebra with constant entries**, (d, x) .

With $E[2]$ irreducible one works instead over the cubic étale algebra $L = \mathbb{Q}[t]/(f(t))$, where $E : Y^2 = f(X)$. The descent map is

$$E(\mathbb{Q}_v)/2 \hookrightarrow (L \otimes \mathbb{Q}_v)^\times / \text{squares}, \quad Q \mapsto X(Q) - \theta,$$

the group of classes is $H^1(\mathbb{Q}, E[2]) = \ker(N_{L/\mathbb{Q}} : L^\times / \text{sq} \rightarrow \mathbb{Q}^\times / \text{sq})$, the local pairing is a product of Hilbert symbols over the places of L ,

$$\langle Q, d \rangle_v = \frac{1}{2} \prod_{w|v} (X(Q) - \theta, d)_w,$$

and the Brauer class is a **corestriction**,

$$\mathcal{A}_d = \text{cor}_{L(E)/\mathbb{Q}(E)}(d, X - \theta).$$

It is still unramified — $\text{div}(X - \theta) = 2(T_\theta) - 2(O)$ over L has even multiplicities, and corestriction preserves that — and still of order 2. It is simply not a quaternion algebra over $\mathbb{Q}(E)$ with constant slots.

5.1. The counterexample, worked

Take the conductor-37 curve in short form,

$$E : Y^2 = X^3 - 16X + 16, \quad (X, Y) = (4x, 8y + 4) \text{ on } y^2 + y = x^3 - x.$$

Its cubic $t^3 - 16t + 16$ is irreducible, so $E[2]$ is irreducible with Galois group S_3 and E has no rational 2-torsion; PARI's `ellisomat` gives isogeny class size 1, so E admits **no rational isogeny of any degree**. And yet the closure of $E(\mathbb{Q})$ in $E(\mathbb{Q}_{23})$ has index 2.

Let $L = \mathbb{Q}(\theta)$ with $\theta^3 = 16\theta - 16$: totally real, discriminant 148, class number 1. Take

$$d = \frac{1}{4}\theta^2 - 2\theta - 11, \quad N_{L/\mathbb{Q}}(d) = 529 = 23^2.$$

The square norm is exactly the condition for d to define a class in $H^1(\mathbb{Q}, E[2])$, and the ideal (d) is the single prime above 23 of residue degree 2 — so d is a unit away from 23, which is what makes all the other places free.

place	$\prod_{w v}$	why
$v = \infty$	+1	checked on every real point
$v = 2, v = 37$	+1	checked on every local point by scanning X
other $v \neq 23$	+1	$v_w(X(Q) - \theta)$ even and d a unit there
$v = 23$	varies	d has odd valuation at the degree-2 prime

Reciprocity over L then forces the $v = 23$ factor to be +1 at every **rational** point — verified on nP for $n \leq 8$, $P = (0, 4)$ the saturated generator, where all four columns and the product are +1. But at $X = -1$ we get $X^3 - 16X + 16 = 31 \equiv 8 \pmod{23}$, a residue, so $E(\mathbb{Q}_{23})$ has a point with $X = -1$, and there

$$\prod_{w|23} (X - \theta, d)_w = -1.$$

No 2-isogeny, no rational 2-torsion, no isogeny at all — and still an index-2 Brauer–Manin obstruction. The implication that **does** hold runs the other way: a rational 2-isogeny makes the class a single quaternion algebra with constant entries, which is the only reason Section 2 and Section 3 look as simple as they do. What governs the index-2 case is 2-descent, and every curve has that.

6. Index 4 and index 11: what is actually harder

Every elliptic curve has n -descent for every n , and the framework is uniform in n : the local conditions $W_v = \text{im}(E(\mathbb{Q}_v)/n)$ are maximal isotropic for the cup product $H^1(\mathbb{Q}_v, E[n])^{\times 2} \rightarrow H^2(\mathbb{Q}_v, \mu_n) = \frac{1}{n}\mathbb{Z}/\mathbb{Z}$, and reciprocity holds. Manin's theorem guarantees the classes exist. So **yes, in principle** — nothing distinguishes $n = 2$ structurally. What separates the cases is explicitness, and it separates them in three stages.

6.1. The order of the class is forced

Both local quotients are **cyclic**:

$$E(\mathbb{Q}_{59})/\Gamma \cong \mathbb{Z}/4, \quad E(\mathbb{Q}_{11})/\Gamma \cong \mathbb{Z}/11.$$

For conductor 37 at $p = 59$: $E(\mathbb{F}_{59}) \cong \mathbb{Z}/52$ is cyclic, $\bar{P}$ has order 13, and $13P \in E_1 \setminus E_2$, so $\Gamma \supseteq E_1$ and the quotient is $\mathbb{Z}/52$ modulo $\mathbb{Z}/13$. For conductor 89 at $p = 11$ the quotient is E_1/E_2 .

A family of classes of order 2 cuts out a quotient of **exponent 2**, so it can never produce $\mathbb{Z}/4$. So index 4 needs a class of order 4 and index 11 one of order 11 — you cannot assemble either from quaternion algebras.

You can, though, get **part** of the index 4. Since $\bar{P}$ has odd order 13 it lies in $2E(\mathbb{F}_{59})$, so 2-descent sees a factor 2:

n	$E(\mathbb{F}_{59})/n$	image of $E(\mathbb{Q})$	index seen
2	$\mathbb{Z}/2$	trivial	2
4	$\mathbb{Z}/4$	trivial	4
13, 26, 52	$\mathbb{Z}/n$	non-trivial	1

A quaternion class gets you halfway; the last factor 2 needs 4-descent.

6.2. The descent algebra grows

For $n = 2$ the descent algebra is the **cubic** étale algebra $L = \mathbb{Q}[t]/(f)$ — which is why Section 5.1 fits on a page and could be done with `bnfinit` on a cubic field. For $n = 11$ on conductor 89 the 11-division polynomial has degree $(11^2 - 1)/2 = 60$ and is **irreducible over $\mathbb{Q}$** , so nothing collapses; $[\mathbb{Q}(E[11]) : \mathbb{Q}]$ is bounded by $|\text{GL}_2(\mathbb{F}_{11})| = 13200$. Explicit n -descent is practical only for small n .

There is also the roots-of-unity bookkeeping: a degree- n cyclic algebra needs μ_n , and $\mu_n \subset \mathbb{Q}$ only for $n = 2$. For $n > 2$ one builds the class over $\mathbb{Q}(\mu_n)$ and corestricts. Tedious, but not an obstruction.

6.3. Wildness at the place where detection happens

This is the real wall. The n -th norm residue symbol at v is **tame** exactly when $v \nmid n$.

case	n	detection place	symbol there
Section 2, Section 5.1	2	7, 23	tame
conductor 37, index 4	4	59	tame ($59 \nmid 4$)
conductor 89, index 11	11	11	wild — $n = p$

For index 4 the symbol is tame **where the class has to be non-trivial**. It is wild at $v = 2$, but there one only needs **vanishing**, and that wildness is the classical mod-8 kind that every system implements. For index 11 the detection place is 11 and $n = 11$: wild exactly where the work has to be done.

And that is structural, not bad luck. $E_1(\mathbb{Q}_p) \cong \mathbb{Z}_p$ is pro- p , so **every** finite quotient of the formal group is a p -group. Cutting $E_1/E_2 \cong \mathbb{Z}/p$ therefore always demands a class of order p , evaluated at the place p — the wild case, every time. By contrast E_0/E_1 has order $p + 1 - a_p$ (good reduction)

or $p \mp 1$ (multiplicative), prime to p except at anomalous primes, so the reduction layer lives in the tame world.

So the filtration $E \supset E_0 \supset E_1 \supset E_2$ of `ec-padic-closure.typ` is **also a tame/wild filtration**: the top layers are tame and explicitly reachable, and the formal-group layer — the one whose failure the naive criterion misses — is exactly the wild one.

6.4. The tooling gap, already met in these notes

PARI’s `nfhilbert` is literally “does $X^2 - aY^2 - bZ^2 = 0$ have a solution”: quadratic only, no exponent argument. Sage’s `hilbert_symbol` is the same. Neither has an n -th norm residue symbol for $n > 2$. `descent-s3.typ` in this directory already hit precisely this wall — it needed a **cubic** norm-residue symbol at the wild place $v = 3$, and recorded that neither system provides one. The index-11 case is the same gap one prime up.

Summary. Index 4 is harder than index 2 only in bookkeeping: the class exists, the symbol is tame where it matters, and 4-descent is implemented in practice. Index 11 is harder in kind: it needs an order-11 class over a torsion field of degree up to 13200, and the symbol that must be non-trivial is the wild 11-th norm residue symbol at 11. Neither case is blocked by mathematics — both are blocked by explicitness, and only the second is blocked by something no system here can compute.

7. Is Brauer–Manin then just descent, repackaged?

For an elliptic curve: essentially yes, and the identification is exact. Three qualifications keep it from being **only** a repackaging.

7.1. Why the identification is exact here

$\overline{E}$ is a curve over an algebraically closed field, so $\text{Br}(\overline{E}) = 0$ by Tsen’s theorem: an elliptic curve has **no transcendental Brauer group at all**. Hence

$$\text{Br}(E) = \text{Br}_1(E), \quad \text{Br}(E)/\text{Br}_0(E) \cong H^1(\mathbb{Q}, E) = \bigcup_n H^1(\mathbb{Q}, E)[n],$$

and $H^1(\mathbb{Q}, E)[n] = \text{coker}(E(\mathbb{Q})/n \rightarrow H^1(\mathbb{Q}, E[n]))$ is precisely the output of n -descent. Pairing against it is precisely the local pairing of n -descent. So the Brauer–Manin condition on E is the intersection over all n of the n -descent reciprocity conditions — there is no class left over, and no obstruction that descent cannot see.

For a fixed (E, p) a single n suffices: $E(\mathbb{Q}_p)/\Gamma$ is a finite abelian group, and n its exponent. Across all p at once you need $n \rightarrow \infty$ along p -powers, because the formal group is $\mathbb{Z}_p$ and only p -power descent reaches its deeper layers.

7.2. Qualification 1: it is a duality, not a restatement

Descent and Brauer–Manin use the same perfect pairing, but read it in opposite directions:

	descent	Brauer–Manin
what is cut down	$H^1(\mathbb{Q}, E[n])$	$\prod_v E(\mathbb{Q}_v)$
what does the cutting	the local conditions W_v	classes $\beta \in H^1(\mathbb{Q}, E)$
what you learn	a bound on $E(\mathbb{Q})/n$, i.e. on the group	a bound on $\overline{E(\mathbb{Q})}$, i.e. on the closure

The sharpening this gives is worth stating, because it says exactly which classes can obstruct at a given place. For $\mathcal{A}_\beta$ to vanish at every $v \neq p$ and every local point means $\beta_v \in W_v^\perp = W_v$ for all $v \neq p$, and to detect at p means $\beta_p \notin W_p$. That is:

The classes that obstruct density at p are exactly the non-trivial elements of

$$\mathrm{Sel}_n^{(p)}/\mathrm{Sel}_n,$$

the Selmer group with its condition at p **relaxed**, modulo the honest Selmer group. The element $d = \frac{1}{4}\theta^2 - 2\theta - 11$ of Section 5.1 is a generator of this quotient for $n = 2$, $p = 23$ on the conductor-37 curve: it satisfies the local conditions at every $v \neq 23$, and it is not in Sel_2 because W_{23} is maximal isotropic, so $W_{23}^\perp = W_{23}$ and pairing non-trivially with W_{23} puts it outside. There $\dim \mathrm{Sel}_2 = 1$ (rank 1, no rational 2-torsion, $\mathrm{III}[2] = 0$) while $\dim \mathrm{Sel}_2^{(23)} \geq 2$.

Whether such a class exists is governed by the Greenberg–Wiles comparison formula — the subject of `selmer-local-conditions.typ` in this directory. So “does the obstruction exist” is not a formality; it is a Selmer-rank computation.

7.3. Qualification 2: that descent sees everything is a theorem

The inclusion $\overline{E(\mathbb{Q})} \subseteq E(\mathbf{A})^{\mathrm{Br}}$ is formal — reciprocity plus continuity. The reverse inclusion, which is what licenses “the obstruction explains all of it”, is Manin’s theorem and needs $\mathrm{III}(E)$ finite; its proof is Poitou–Tate exactness and the Cassels–Tate pairing. Without finiteness of III you do not know that the n -descents, however many you run, detect every adelic point outside the closure. So the completeness of the repackaging rests on an open conjecture in general, and on Gross–Zagier and Kolyvagin for the analytic-rank- ≤ 1 curves treated here.

7.4. Qualification 3: the equivalence is special to E

Off an abelian variety there is no group law, no isogeny, and no n -descent — but there is still a Brauer group and still $\sum_v \mathrm{inv}_v = 0$. That is exactly the situation of the other two documents in this directory: the class $(2, x + 4)$ on the negative Pell conic of `integral-bm.typ` is not an n -descent class of anything. The Brauer formulation buys **portability**, and on E specifically it buys nothing new — which is the observation you started from.

So: on an elliptic curve, Brauer–Manin = all the n -descents together, exactly and with nothing left over, because $\mathrm{Br}(\overline{E}) = 0$. It is a repackaging — but a dual one, which turns “bound the Mordell–Weil group” into “bound its closure”; its completeness is Manin’s theorem, not bookkeeping; and the packaging is what lets the same argument run on varieties that are not groups.

8. Relation to the other two documents

- `ec-padic-closure.typ` computes **that** the closure is a proper open subgroup, and **which** layer of $E \supset E_0 \supset E_1 \supset E_2$ fails, by a filtration index $(N_p/m)p^{v-1}$. It is a statement about $E(\mathbb{Q}_p)$ and needs no global input.
- This document explains **why**, globally: reciprocity applied to one class, exactly as in `integral-bm.typ`. The two obstructions even have the same shape — there $(2, x + 4)$ on the negative Pell conic, here (d, x) on an elliptic curve, both unramified because the relevant function has even divisor, both evaluated by Hilbert symbols, both concluded by $\sum_v \mathrm{inv}_v = 0$.
- The difference is what is being obstructed. In `integral-bm.typ` the obstruction empties an **entire** adelic space and proves $\mathcal{X}(\mathbb{Z}) = \emptyset$. Here it empties an open **subset** of $E(\mathbb{Q}_p)$, which is a weaker conclusion — $E(\mathbb{Q})$ is large — but the same mechanism.

Companion file: `ec-density-bm.gp`, run as

```
gp -q -s 40000000000 ec-density-bm.gp < /dev/null > results/ec-density-bm.txt
```

It verifies the ranks and analytic ranks, the local images S_v at $v = 2$ (by scanning rational x with $x(x^2 + ax + b)$ a 2-adic square), the invariant of $\mathcal{A}_d$ at every place on many rational points together with the reciprocity product, the local point at p where the invariant is $1/2$, and the index table of Section 4.